

365 Data Science SQL Basics – Topic 1.1: Data Definition Language (DDL)

Exhaustive Notes on DDL Statements: CREATE, ALTER, DROP, TRUNCATE & Schema Management

1. Data Definition Language (DDL) Overview

Data Definition Language (DDL) consists of SQL statements used to define, alter, and delete database structures such as tables, schemas, views, and indexes. DDL commands directly alter the relational blueprint maintained in the system catalog.

SQL Sub-Languages Taxonomy (DDL, DML, DCL, TCL, DQL)

DDL (Data Definition)
CREATE, ALTER, DROP, TRUNCATE
Modifies database schema structure

DML (Data Manipulation)
INSERT, UPDATE, DELETE
Modifies data rows inside tables

DCL & TCL
DCL: GRANT, REVOKE
TCL: COMMIT, ROLLBACK

DQL (Data Query)
SELECT statement
Queries and filters records

Diagram: SQL Sub-Languages Taxonomy (DDL, DML, DCL, TCL, DQL).

2. The CREATE Statement

The `CREATE` statement instantiates new database objects. When creating a table, column names, data types, and integrity constraints must be specified.

```
CREATE TABLE sales (  
    purchase_number INT AUTO_INCREMENT PRIMARY KEY,  
    date_of_purchase DATE NOT NULL,  
    customer_id INT,  
    item_code VARCHAR(10)  
);
```

3. The ALTER Statement

The `ALTER` statement modifies existing table structures by adding, dropping, or altering columns and constraints.

```
-- Adding a foreign key constraint to sales table  
ALTER TABLE sales  
ADD CONSTRAINT fk_sales_customers  
FOREIGN KEY (customer_id) REFERENCES customers(customer_id)  
ON DELETE CASCADE;
```

4. The DROP and TRUNCATE Statements

- `TRUNCATE`: Removes all data rows from a table while preserving the table schema and resetting auto-increment counters.
- `DROP`: Permanently deletes the table structure, schema, and all data from the database.

5. Topic 1.1 Viva & Interview Q&A

1. Q: Does TRUNCATE delete the table structure? A: No. TRUNCATE deletes all data rows while keeping the empty table structure intact. 2. Q: Why use ALTER TABLE instead of recreating the table? A: ALTER TABLE modifies existing table definitions without losing currently stored data rows.